Lesson plan · for the teacher

Joining tables for reports

Data engineering · Level 3 · Lesson 4

Time
40 min

Materials
Column cards: name, class, marks, attendance, phone, address, health notes, date of birth; Board and chalk; Optional: a phone or computer for the SQL steps

Learning goals

- Join tables with JOIN … ON.
- Choose an inner or a left join, and find who's missing.
- Write reports that show only what each reader needs.

Activity: A report for one reader

1. Warm-up (5 min): List on the board the reports the school makes and who reads each one.
2. Teach (8 min): With two small tables on the board, show an inner join (Zara vanishes) and a left join (Zara with an empty mark).
3. Sort (12 min): Groups sort the column cards into what a parent, the nurse and the head need, and put aside the cards nobody needs.
4. Write (10 min): Each group writes the SELECT for one reader's report, with only that reader's columns and rows.
5. Wrap-up (5 min): Ask which columns nobody needed, and whether the school should keep collecting them. Then give the exit ticket.

Use made-up tables on the board. The discussion of health notes can be sensitive: keep it general, about kinds of data, not about any real pupil.

Discussion

- Why did Zara vanish from the inner join?
- How would you find every pupil who missed a test?
- Which columns should never be in a marks report?

Exit ticket

- Write the SELECT and FROM lines of a report for parents, and say which join you'd use and why.

Try it at school tomorrow
Ask your pupils what a parent, the school nurse and the head teacher each need from the school records. They sort column cards into three piles. Which columns does nobody need at all?

Practice questions · for you

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1 **A report for parents' evening goes to each family. What should it show?**

- ◯ Their child's name and their child's marks
- ◯ Name, class, every mark, phone and health notes
- ◯ Every pupil's name and marks

2 **Show each pupil's name next to their maths mark from yesterday's test, by joining pupils to marks on the pupil's ID.**

```
SELECT name, mark FROM pupils
JOIN marks;
```

Write your program:

3 **A LEFT JOIN keeps every row of the first table, with NULL where there is no match.**

- ◯ True
- ◯ False

4 **Who missed yesterday's maths test? Show the names of the pupils who have no mark.**

```
SELECT name FROM pupils
JOIN marks ON marks.pupil_id = pupils.id
WHERE mark IS NULL;
```

Write your program:

5 **A report for the head teacher: each class's average mark, rounded to 1 decimal place, in class order. Put the lines in order, leaving out the one the head doesn't need.**

Number the lines in order (1, 2, 3…). Cross out any line that doesn't belong.

- ☐ `FROM pupils`
- ☐ `GROUP BY pupils.class`
- ☐ `SELECT pupils.name, pupils.class, marks.mark`
- ☐ `ORDER BY pupils.class;`
- ☐ `SELECT pupils.class, ROUND(AVG(marks.mark), 1)`
- ☐ `JOIN marks ON marks.pupil_id = pupils.id`

✎ **Pick one report your school makes. Who reads it, and which columns could it leave out?**

Check your answers

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1 **Their child's name and their child's marks**

Each family needs its own child's name and marks, and nothing about other children.

2

```
SELECT name, mark FROM pupils
JOIN marks ON marks.pupil_id = pupils.id;
```

ON says how rows match: a mark belongs to the pupil whose id equals its pupil_id.

3 **True**

That's the difference from an inner join: no pupil vanishes.

4

```
SELECT name FROM pupils
LEFT JOIN marks ON marks.pupil_id = pupils.id
WHERE mark IS NULL;
```

The inner join keeps only pupils with a mark, so none of them has a NULL mark: no rows.

5 **Top to bottom: 2, 4, ✗, 5, 1, 3**

```
SELECT pupils.class, ROUND(AVG(marks.mark), 1)
FROM pupils
JOIN marks ON marks.pupil_id = pupils.id
GROUP BY pupils.class
ORDER BY pupils.class;
```

SQL needs its clauses in this order: SELECT, FROM, JOIN … ON, GROUP BY, ORDER BY.